

SmartClass Vision — Quick Reference Summary

Enterprise AI Biometric Attendance & Security Architecture | High-Yield Cheat Sheet

1. Core AI Models & Why We Use Them

Model	Role in System	Key Benefit & Why Selected
OpenCV YuNet (2023 ONNX)	Face Detection & 5-Point Landmarks	Ultra-fast C++ DNN (~15ms CPU). Detects micro-faces & extracts 5 key landmarks (eyes, nose, mouth) for affine alignment .
InsightFace ArcFace (ResNet-50 / 512-D)	Biometric Face Recognition	Additive Angular Margin Loss maximizes inter-class distance. Outperforms FaceNet/dlib with 99.8% accuracy on normalized 512-D vectors.
Liveness & Quality Engine	Anti-Spoofing & Quality Filter	Rejects motion blur, extreme head turns (yaw > 1.4), dark shots, and screen photo attacks using Moiré & frequency analysis.

2. Key Thresholds & Numbers to Remember (Quick Cheat Sheet)

Metric	Value	Purpose & Practical Meaning
Cosine Similarity	≥ 0.65	ArcFace confidence threshold required to confirm match.
Top-1 vs Top-2 Margin	≥ 0.12	Ambiguity rejection gate. Prevents misidentifying lookalikes/twins.
Burst Capture Size	10 Shots	Synchronized multi-shot capture sequence across the classroom.
Consensus Requirement	≥ 5 of 10	Student must appear in ≥ 5 shots to eliminate accidental passers-by.
Min Sharpness (Laplacian)	40.0	Discards blurry frames or moving faces.
Registration Poses	5 Angles	Center, Left, Right, Up, Down for complete facial profile.

3. Summary of Recent Major Updates

- **Complete ArcFace + YuNet Migration:** Removed legacy FaceNet & YOLO models; standardized on 512-D L2 embeddings.
- **Strict Section & Roster Validation:** Students are validated against enrolled section. Cross-class students are flagged *REJECTED - WRONG CLASS*.
- **Cryptographic HMAC-SHA256 Signatures:** Every attendance CSV is signed with file hash + faculty ID to prevent manual Excel edits.
- **Automated Audit Logs:** Rejections, spoof attempts, and mismatch reasons are automatically logged to SQLite database.
- **Optimized Video Streaming:** Real-time live HUD overlay with instant countdowns, enrolled candidate counts, and green capture feedback.

SmartClass Vision • Designed for high-speed, secure, multi-student biometric classroom attendance.